9. (i) Solve, for $0 \leqslant \theta < \pi$, the equation

$$\sin 3\theta - \sqrt{3}\cos 3\theta = 0$$

giving your answers in terms of π

(3)

(ii) Given that

$$4\sin^2 x + \cos x = 4 - k, \quad 0 \leqslant k \leqslant 3$$

(a) find $\cos x$ in terms of k

(3)

(b) When $k = 3$, find the values of x in the range $0 \leqslant x < 360°$

(3)